ENGINEERING CHEMISTRY (MODULE NOTES)

Complete Technical & Examination Study Guide — HBTU Kanpur B.Tech Syllabus

Core Units Covered: Electrochemistry • Nanochemistry • Environmental Chemistry
Scope: Complete Theoretical Derivations, Cell Reactions, Analytical Protocols & Control Tech

Target: End-Semester University Exams & Competitive Prep (GATE / PSUs)
Specification: Strictly Exhaustive — Zero Omission Standard

PART 1 ELECTROCHEMISTRY & ENERGY STORAGE DEVICES

Electrochemistry is the branch of chemical science dealing with the interconversion of chemical energy and electrical energy through oxidation-reduction (redox) reactions occurring at phase boundaries (electrodes).

1. Electrochemical Cells: Principles & Fundamentals

An electrochemical cell consists of two half-cells (electrodes) immersed in an electrolyte, connected internally by a porous membrane or salt bridge and externally by an electrical circuit.

- **Galvanic (Voltaic) Cell:** Converts spontaneous chemical reaction energy ($\Delta G < 0$) into electrical energy (e.g., Daniell Cell).
- **Electrolytic Cell:** Uses external electrical energy to drive a non-spontaneous chemical change ($\Delta G > 0$).

Daniell Cell Construction & Convention

- **Anode (Oxidation):** Zinc rod immersed in $ZnSO_4(aq)$. Zinc dissolves, releasing electrons (negative polarity in galvanic cell).
- **Cathode (Reduction):** Copper rod immersed in $CuSO_4(aq)$. Copper ions gain electrons and plate out (positive polarity).
- **Salt Bridge:** Inverted U-tube containing an agar-agar gel soaked in inert electrolyte (KCl, KNO_3, NH_4NO_3). Maintains electrical neutrality and eliminates liquid junction potential.

Cell Representation & Nernst Equation

Standard IUPAC representation:

$Zn(s) \mid Zn^{2+}(aq, C_1) \parallel Cu^{2+}(aq, C_2) \mid Cu(s)$

For a general redox process: $aA + bB \rightleftharpoons cC + dD$:

$E_{cell} = E°_{cell} - (RT / nF) \cdot \ln(Q)$

At 298 K (25°C):

$E_{cell} = E°_{cell} - (0.0591 / n) \cdot \log_{10}([C]^c[D]^d / [A]^a[B]^b)$

DANIELL CELL HALF-CELL & NET REDOX REACTIONS

Anode (Oxidation): $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ $[E°_{ox} = +0.76\ V]$

Cathode (Reduction): $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ $[E°_{red} = +0.34\ V]$

Net Overall Cell Reaction: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ $[E°_{cell} = E°_{cathode} - E°_{anode} = 0.34 - (-0.76) = 1.10\ V]$

2. Primary Battery: Dry Cell (Leclanché Cell & Alkaline Cell)

A dry cell is a primary electrochemical cell in which the electrolyte is immobilised as a moist paste, preventing liquid leakage and making it portable.

A. Standard Leclanché Dry Cell

- **Anode:** Zinc metal cylindrical container / outer can, which also serves as the container: $Zn(s)$.
- **Cathode:** Centrally placed inert graphite (carbon) rod surrounded by a powdered mixture of manganese dioxide (MnO_2) and carbon black (to enhance electrical conductivity).
- **Electrolyte:** Moist paste of ammonium chloride (NH_4Cl) and zinc chloride ($ZnCl_2$) with starch/flour as binder.

DRY CELL CHEMICAL REACTIONS

Anode (Oxidation): $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$

Cathode (Reduction): $2MnO_2(s) + 2NH_4^+(aq) + 2e^- \rightarrow Mn_2O_3(s) + 2NH_3(g) + H_2O(l)$ (or formation of $MnO(OH)$)

Complexation Reaction (Preventing gas pressure buildup): Ammonia gas is trapped by Zn^{2+} to form a stable complex diamminedichlorozinc(II):

$Zn^{2+} + 2NH_3 + 2Cl^- \rightarrow [Zn(NH_3)_2Cl_2](s)$

Cell Potential: Approximately **1.5 V**.

Limitations: Acidic nature of NH_4Cl slowly corrodes the zinc container even when not in use (shelf-life degradation); voltage drops significantly under high continuous discharge.

B. Alkaline Dry Cell (Improvement over Leclanché)

Employs potassium hydroxide (KOH) as the alkaline electrolyte instead of acidic NH_4Cl. The anode consists of granulated zinc powder dispersed in a gel (higher surface area), and the cathode is compressed MnO_2.

- **Anode Reaction:** $Zn(s) + 2OH^-(aq) \rightarrow ZnO(s) + H_2O(l) + 2e^-$
- **Cathode Reaction:** $2MnO_2(s) + H_2O(l) + 2e^- \rightarrow Mn_2O_3(s) + 2OH^-(aq)$
- **Advantages:** Zinc does not corrode readily in basic media; maintains a steady 1.5 V under load; operates efficiently at low temperatures; higher energy density.

3. Fuel Cells (Hydrogen-Oxygen Fuel Cell / Bacon Cell)

A **fuel cell** is an electrochemical galvanic device that continuously converts the chemical energy of a continuously supplied fuel (such as hydrogen, natural gas, methanol) and an oxidant directly into electrical energy without undergoing thermal combustion.

Construction Details

• **Electrodes:** Porous graphite rods impregnated with finely dispersed catalysts (Pt, Pd, or Ag) to facilitate electrode kinetics.

• **Electrolyte:** Concentrated hot aqueous solution of KOH (25–40%) operating at ~100–200°C, or a Proton Exchange Membrane (PEMFC).

• **Feed Gases:** H_2 fuel continuously fed into anode compartment; O_2 (or purified air) fed into cathode compartment.

Electrochemical Efficiency

Unlike thermal power plants limited by the Carnot cycle efficiency ($\eta_{Carnot} = 1 - T_C/T_H \approx 35–40\%$), fuel cells are not heat engines.

Theoretical efficiency:

$\eta_{max} = (\Delta G / \Delta H) \times 100\% \approx 83\%$ at 298 K.

Practical operating efficiency typically reaches **60–70%** (and up to 85% in combined heat and power systems).

H_2-O_2 ALKALINE FUEL CELL ELECTRODE REACTIONS

Anode (Oxidation): $2H_2(g) + 4OH^-(aq) \rightarrow 4H_2O(l) + 4e^-$ $[E°_{ox} = +0.83\ V]$

Cathode (Reduction): $O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$ $[E°_{red} = +0.40\ V]$

Overall Cell Reaction: $2H_2(g) + O_2(g) \rightarrow 2H_2O(l)$ $[E°_{cell} = 1.229\ V \approx 1.23\ V]$

Salient Advantages: Zero harmful emissions (pure water is the only byproduct, used by astronauts in Apollo space missions); vibration-free and silent operation; modular design; high energy-to-weight ratio.

Engineering Challenges: High cost of noble metal catalysts (Pt); storage, transport, and embrittlement issues of pure hydrogen gas; electrolyte carbonation by ambient CO_2 ($2KOH + CO_2 \rightarrow K_2CO_3 + H_2O$) which blocks catalyst pores.

4. Photovoltaic Cells & Solar Cells

A **photovoltaic (PV) cell** is a specialized semiconductor solid-state diode that converts photon radiant energy directly into electrical energy via the photovoltaic effect.

A. Working Principle of p-n Junction Solar Cell

1. **Absorption of Light & Generation of Electron-Hole (e^--h^+) Pairs:** Incident solar photons with energy $h\nu \geq E_g$ (where E_g is semiconductor bandgap, ~1.12 eV for crystalline silicon) excite electrons from the valence band to the conduction band, creating electron-hole pairs in the depletion region.
2. **Charge Carrier Separation:** The built-in electric field across the p-n junction rapidly sweeps photogenerated electrons toward the n-type layer and holes toward the p-type layer before recombination can occur.
3. **Current Collection:** The separated carriers are collected by metallic contacts (front silver grid and continuous back contact), establishing a potential difference (V_{oc}) and driving a direct current (I_{sc}) through an external electrical load.

Key Parameters:

Open-Circuit Voltage (V_{oc}) | Short-Circuit Current (I_{sc})

Fill Factor (FF): $FF = (P_{max}) / (V_{oc} \times I_{sc}) = (V_{mp} \times I_{mp}) / (V_{oc} \times I_{sc})$ (typically 0.7 to 0.85)

Photovoltaic Conversion Efficiency: $\eta = (P_{max} / P_{in}) \times 100\% = (V_{oc} \times I_{sc} \times FF / [G \times A]) \times 100\%$

5. Dye-Sensitized Solar Cells (DSSC / Grätzel Cell)

A **Dye-Sensitized Solar Cell (DSSC)**, pioneered by Michael Grätzel in 1991, is a thin-film photoelectrochemical solar cell inspired by biological photosynthesis. Unlike conventional p-n junction cells where the semiconductor performs both light absorption and charge transport, DSSC separates these functions.

Architecture & Components of a DSSC:

- **Transparent Conducting Substrate:** Fluorine-doped Tin Oxide (FTO) coated glass plates.
- **Photoanode (Mesoporous Semiconductor):** Nanoporous titanium dioxide (TiO_2, anatase phase, particle size ~20 nm) sintered onto FTO. Provides enormous internal surface area (~1000× flat substrate).
- **Sensitizing Dye (Chromophore):** Monolayer of transition metal coordination complex (e.g., Ruthenium polypyridyl complexes like N3, N719 dye) chemically anchored to the TiO_2 surface.
- **Redox Electrolyte:** Organic solvent containing iodide/triiodide (I^- / I_3^-) redox couple.
- **Counter Electrode (Cathode):** FTO glass coated with a thin catalytic layer of Platinum (Pt) or carbon.

DSSC STEP-BY-STEP PHOTOCYCLE MECHANISM

1. **Photoexcitation of Dye:** $Dye + h\nu \rightarrow Dye^*$ (Electron promoted to LUMO of dye)
2. **Electron Injection:** $Dye^* + TiO_2 \rightarrow Dye^+ + e^-(CB\ of\ TiO_2)$ (Ultrafast injection in femtoseconds)
3. **Diffusion & External Work:** Electrons percolate through mesoporous TiO_2 network, enter FTO, travel through external load to counter electrode.
4. **Dye Regeneration by Electrolyte:** $2Dye^+ + 3I^- \rightarrow 2Dye + I_3^-$
5. **Electrolyte Regeneration at Cathode:** $I_3^- + 2e^-(from\ external\ circuit) \rightarrow 3I^-$ (Catalyzed by Pt)

Net Result: Light is converted to electric power with zero chemical degradation of components!

DSSC Advantages vs Silicon PV: Low fabrication cost (roll-to-roll printing without high vacuum or ultra-clean rooms); excellent diffuse light and indoor light harvesting; flexible and semitransparent; lower temperature coefficient.

PART 2 NANOCHEMISTRY & ADVANCED NANOMATERIALS

1. Introduction to Nanochemistry & Nanoscale Phenomena

Nanochemistry is the synthesis, characterization, and application of chemical building blocks and structures within the nanoscale dimension of **1 to 100 nanometres (1 nm = 10^{-9} m)**. At this scale, physical and chemical properties deviate dramatically from classical bulk behaviour due to two primary phenomena:

- **Quantum Confinement Effect:** When the physical dimension of a crystal becomes comparable to or smaller than the de Broglie wavelength or exciton Bohr radius of charge carriers, the electronic energy bands split into discrete, quantized energy levels. The bandgap increases as particle size decreases ($E_g \propto 1/r^2$), causing blue-shifts in optical absorption/photoluminescence (e.g., CdSe quantum dots changing color from red to blue as diameter decreases from 6 nm to 2 nm).
- **High Specific Surface Area-to-Volume Ratio:** As spherical radius r decreases, surface area per unit volume scales as $3/r$. A high percentage of atoms reside directly on the surface with unsatisfied coordination numbers (dangling bonds), leading to hyper-reactivity, depressed melting points (Gibbs-Thomson effect), and extraordinary catalytic activity.

2. General Methods of Synthesis of Nanomaterials

Nanomaterial fabrication routes are categorized into two fundamental paradigms: **Top-Down** and **Bottom-Up**.

Synthesis Paradigm	Underlying Principle	Common Methodologies	Characteristics
Top-Down Approach	Physical or mechanical breakdown of bulk macroscopic materials into nanoscale fragments.	• High-Energy Ball Milling • Photolithography & Electron-Beam Lithography • Laser Ablation & Sputtering	High throughput; structural defects and surface contamination often present; broad size distribution.
Bottom-Up Approach	Building nanostructures atom-by-atom, molecule-by-molecule through chemical assembly and nucleation.	• Sol-Gel Process • Chemical Vapor Deposition (CVD) • Chemical Reduction Method • Hydrothermal / Solvothermal Synthesis	Precise control over particle morphology, size, and monodispersity; atomic-level purity.

Detailed Synthesis Mechanisms:

- **Sol-Gel Method:** A wet-chemical synthesis involving hydrolysis and polycondensation of metal alkoxide precursors (e.g., Tetraethyl orthosilicate - TEOS, $Si(OC_2H_5)_4$ or Titanium isopropoxide).

> **Hydrolysis:** $M\text{-}OR + H_2O \rightarrow M\text{-}OH + R\text{-}OH$
> **Condensation:** $M\text{-}OH + M\text{-}OH \rightarrow M\text{-}O\text{-}M + H_2O$ or $M\text{-}OH + M\text{-}OR \rightarrow M\text{-}O\text{-}M + R\text{-}OH$
> A colloidal suspension (sol) transitions into a 3D interconnected network (gel), followed by aging, drying (forming xerogel or aerogel), and calcination to yield uniform metal oxide nanoparticles.

- **Chemical Vapor Deposition (CVD):** Volatile gaseous precursors are thermally cracked or catalytically reacted in a high-temperature furnace (500–1000°C) over catalytic transition metal nanoparticles (Fe, Co, Ni). Widely used to produce high-purity Carbon Nanotubes (CNTs) and graphene sheets.

$$CH_4(g) \xrightarrow{\text{[Ni catalyst, 800°C]}} C(\text{nanotube/graphene}) + 2H_2(g)$$

- **Chemical Reduction (Colloidal Gold/Silver Synthesis):** Reduction of metal salts in solution using reducing agents (e.g., sodium borohydride $NaBH_4$ or trisodium citrate) in the presence of capping ligands (e.g., PVP, CTAB) to prevent agglomeration by steric or electrostatic stabilization.

$$2HAuCl_4 + 3C_6H_8O_7\ (\text{Citric Acid}) \rightarrow 2Au°\ (\text{nanoparticles}) + 3C_5H_6O_5 + 8HCl + 3CO_2$$

3. Classification of Nanomaterials

Nanomaterials are systematically classified based on the number of dimensions falling outside the 1–100 nm scale (confinement dimensions):

Class	Dimensions < 100 nm	Non-confined Dimensions	Prominent Examples
Zero-Dimensional (0D)	All 3 dimensions < 100 nm	0	Quantum dots (CdSe, PbS), nanoparticles (Au, Ag), fullerenes (C_{60}), nanoclusters.
One-Dimensional (1D)	2 dimensions < 100 nm	1 dimension (> 100 nm)	Nanotubes (CNTs), nanowires (ZnO, Si), nanorods, nanofibres.
Two-Dimensional (2D)	1 dimension < 100 nm (thickness)	2 dimensions (> 100 nm)	Graphene, MXenes, MoS_2 monolayers, nanocoatings, thin films.
Three-Dimensional (3D)	0 dimensions < 100 nm (bulk)	All 3 dimensions bulk	Polycrystalline compacts with nanograins, nanocomposites, dendrimers.

Carbon Allotrope Nanomaterials:

- **Fullerenes (C_{60}, Buckyball):** 60 carbon atoms arranged in a truncated icosahedron (cage of 12 pentagons and 20 hexagons) with sp^2 hybridization. High electron affinity; acts as a free radical scavenger.
- **Carbon Nanotubes (CNTs):** Seamless rolled cylinders of graphene sheets.
  - *Single-Walled (SWCNT):* Single cylinder, diameter 0.8–2 nm. Chirality vector $C_h = na_1 + ma_2$ determines metallic ($n - m = 3q$) or semiconducting nature.
  - *Multi-Walled (MWCNT):* Concentric nested tubes separated by interlayer spacing of ~0.34 nm. Tensile strength ~100× steel; thermal conductivity ~3000 W/m·K.

- **Graphene:** Single 2D atomic monolayer of sp^2 hybridized carbon atoms arranged in a honeycomb lattice. Zero-bandgap semimetal with relativistic Dirac fermions, extraordinary electron mobility (~200,000 $cm^2/V \cdot s$), optical transparency (97.7%), and Young's modulus of 1 TPa.

4. Engineering & Scientific Applications of Nanomaterials

- **Medicine & Targeted Drug Delivery:** Functionalized magnetic nanoparticles (Fe_3O_4) guided by external magnetic fields for hyperthermia cancer therapy; liposomes and polymeric nanocarriers for targeted chemotherapeutic drug delivery (reducing systemic cytotoxicity); gold nanoparticles for photothermal therapy and surface plasmon resonance (SPR) bio-sensing.
- **Catalysis & Chemical Processing:** Nano-sized transition metal catalysts (Pt, Pd, Au nanoparticles) provide massive active surface sites for low-temperature oxidation, hydrogenation, and water splitting for clean hydrogen fuel.
- **Energy Storage & Conversion:** Nanostructured lithium iron phosphate ($LiFePO_4$) and silicon nanowire anodes in Li-ion batteries drastically reduce lithium-ion diffusion distance, enabling ultra-fast charging; carbon nanotube / graphene supercapacitors with ultra-high power densities.
- **Environmental Remediation:** Nanoscale zero-valent iron (nZVI) for in-situ dechlorination of organic solvents and heavy metal immobilization in contaminated aquifers; photocatalytic degradation of organic dyes using TiO_2 and ZnO under UV/solar irradiation.

PART 3 ENVIRONMENTAL CHEMISTRY & POLLUTION CONTROL

1. Air & Water Pollution: Fundamentals & Classifications

Air Pollution: Introduction of deleterious solid, liquid, or gaseous agents into the atmosphere at concentrations exceeding ambient tolerance thresholds.

- **Primary Air Pollutants:** Emitted directly from identifiable sources (e.g., CO, SO_2, NO, NO_2, H_2S, particulate matter).
- **Secondary Air Pollutants:** Formed in the atmosphere via chemical interactions among primary pollutants and solar irradiation (e.g., O_3, PAN (Peroxyacetyl nitrate), photochemical smog, H_2SO_4, HNO_3).

Water Pollution: Physical, chemical, or biological alteration of water quality rendering it toxic or unusable. Key categories: pathogenic organisms, oxygen-demanding organic wastes, plant nutrients (nitrogen and phosphorus causing *eutrophication*), toxic heavy metals (Pb, Cd, Hg, As, Cr), and synthetic organic chemicals (pesticides, detergents, endocrine disruptors).

2. Chemical Analysis of Gaseous Effluents

A. Oxides of Sulphur (SO_x / SO_2) Analysis — West-Gaeke Spectrophotometric Method

Principle: Flue gas or air containing sulphur dioxide (SO_2) is bubbled through a trapping solution of 0.04 M potassium or sodium tetrachloromercurate(II) ($K_2[HgCl_4]$). A highly stable, non-volatile dichlorosulphitomercurate complex is formed:

$SO_2 + H_2O + [HgCl_4]^{2-} \rightarrow [HgCl_2(SO_3)]^{2-} + 2H^+ + 2Cl^-$

The complex is then reacted with formaldehyde ($HCHO$) and bleached pararosaniline hydrochloride (in acid solution). It forms an intensely colored magenta-purple pararosaniline methylsulphonic acid dye.

Absorbance is measured spectrophotometrically at **λ = 548 nm**. Beer-Lambert's law ($A = \varepsilon \cdot c \cdot l$) is applied to determine precise SO_2 concentration against a standard calibration curve.

B. Oxides of Nitrogen (NO_x: NO + NO_2) Analysis — Jacobs-Hochheiser / Saltzman Method

Principle: Ambient air is drawn through an aqueous absorbing solution containing sodium hydroxide and sodium arsenite (or acetic acid containing sulfanilic acid and N-(1-naphthyl)-ethylenediamine dihydrochloride - NEDA).

1. Nitric oxide (NO) is pre-oxidized to nitrogen dioxide (NO_2) using solid CrO_3 or $KMnO_4$.
2. NO_2 reacts with sulfanilic acid in an acidic medium via diazotization to form a diazonium cation:

$NH_2\text{-}C_6H_4\text{-}SO_3H + NO_2^- + 2H^+ \rightarrow [^+N{\equiv}N\text{-}C_6H_4\text{-}SO_3H] + 2H_2O$

3. The diazonium salt couples with NEDA to form an azo dye with deep reddish-purple coloration.

Color intensity is measured spectrophotometrically at **λ = 540 nm**. (In continuous industrial emission monitoring, gas-phase *Chemiluminescence* with ozone ($NO + O_3 \rightarrow NO_2{}^* + O_2 \rightarrow NO_2 + h\nu$) at 600–875 nm is standard).

C. Hydrogen Sulphide (H_2S) Analysis — Methylene Blue Spectrophotometric Method

Principle: H_2S is scrubbed from the gas stream by bubbling through an alkaline suspension of cadmium hydroxide or zinc acetate, precipitating cadmium sulfide (CdS) or zinc sulfide (ZnS), which fixes the volatile sulfide.

The precipitated sulfide is reacted in an acidic medium with N,N-dimethyl-p-phenylenediamine sulfate in the presence of iron(III) chloride ($FeCl_3$) catalyst:

$2\ (CH_3)_2N\text{-}C_6H_4\text{-}NH_2 + H_2S + 6Fe^{3+} \rightarrow \text{Methylene Blue Dye (cation)} + 6Fe^{2+} + NH_4^+ + 4H^+$

The resulting methylene blue complex is measured colorimetrically at **λ = 670 nm**.

3. Chemical Analysis of Effluent Liquid Streams: BOD, COD & DO

A. Dissolved Oxygen (DO) — Winkler's Iodometric Titration Method

Dissolved oxygen is fundamental to aquatic aerobic respiration. Its saturation level at 25°C is ~8.3 mg/L.

WINKLER METHOD CHEMICAL CASCADE

1. **Precipitation:** Water sample is treated with $MnSO_4$ and alkaline-iodide-azide reagent ($NaOH + KI + NaN_3$). DO oxidizes white $Mn(OH)_2$ to brown basic manganese oxide:

$Mn^{2+} + 2OH^- \rightarrow Mn(OH)_2(s)$ [white precipitate]

$2Mn(OH)_2(s) + O_2 \rightarrow 2MnO(OH)_2(s)$ [brown precipitate]

2. **Acidification & Iodine Liberation:** Concentrated H_2SO_4 dissolves the brown precipitate, liberating free iodine equivalent to original DO:

$MnO(OH)_2(s) + 2I^- + 4H^+ \rightarrow Mn^{2+} + I_2 + 3H_2O$

3. **Titration:** Liberated I_2 is titrated against standard sodium thiosulfate ($0.025\ N\ Na_2S_2O_3$) using freshly prepared starch indicator (blue to colorless endpoint):

$I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$ (tetrathionate ion)

$$DO\ (mg/L\ or\ ppm) = (V_{titrant} \times N_{thiosulfate} \times 8 \times 1000) / V_{sample}$$

B. Biochemical Oxygen Demand (BOD)

Definition: BOD is the amount of dissolved oxygen required by aerobic microorganisms to biologically oxidize and stabilize biodegradable organic matter in a unit volume of wastewater over a specified incubation period (standard: **5 days at 20°C**, denoted as BOD_5).

- **Test Procedure:** Wastewater is diluted with aerated nutrient-buffered dilution water (seeded with bacteria if necessary). DO of one aliquot is measured immediately (DO_0). The remaining aliquot is sealed in an airtight glass stoppered BOD bottle and incubated in the dark at 20 ± 1°C for 5 days. Final dissolved oxygen (DO_5) is measured.

$$BOD_5\ (mg/L) = [(DO_0 - DO_5) - (B_0 - B_5)(1 - 1/P)] \times P$$

Where: DO_0, DO_5 = Initial & 5-day DO of diluted sample; B_0, B_5 = Initial & 5-day DO of blank; P = Dilution factor (V_{total} / V_{sample}).

C. Chemical Oxygen Demand (COD)

Definition: COD is the total amount of oxygen consumed to chemically oxidize both biodegradable and non-biodegradable organic compounds in wastewater using a powerful chemical oxidizing agent under boiling acidic conditions.

- **Reagent System:** Standard potassium dichromate ($0.25\ N\ K_2Cr_2O_7$) in concentrated H_2SO_4, catalyzed by silver sulfate (Ag_2SO_4 to oxidize straight-chain aliphatic hydrocarbons), with mercuric sulfate ($HgSO_4$ added to complex chloride ions and prevent interference: $6Cl^- + Cr_2O_7^{2-} + 14H^+ \rightarrow 3Cl_2 + 2Cr^{3+} + 7H_2O$).
- **Digestion & Titration:** Refluxed in an open or closed reflux digestion mantle for 2 hours at 150°C. Unreacted residual $Cr_2O_7^{2-}$ is back-titrated with standard ferrous ammonium sulfate [FAS, $Fe(NH_4)_2(SO_4)_2{\cdot}6H_2O$] using ferroin indicator (color change from bluish-green to distinct reddish-brown). A blank with pure distilled water is treated identically.

COD CHEMICAL DIGESTION & TITRATION REACTIONS

Organic Oxidation: $C_nH_aO_bN_c + d\ Cr_2O_7^{2-} + (8d+c)\ H^+ \rightarrow n\ CO_2 + (a+8d-3c)/2\ H_2O + c\ NH_4^+ + 2d\ Cr^{3+}$

Back-titration with FAS: $Cr_2O_7^{2-} + 14H^+ + 6Fe^{2+} \rightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$

$$COD\ (mg/L\ or\ ppm) = [(V_{blank} - V_{sample}) \times N_{FAS} \times 8 \times 1000] / V_{sample\ (mL)}$$

Analytical Parameter	Biochemical Oxygen Demand (BOD)	Chemical Oxygen Demand (COD)
Target Organic Fraction	Biodegradable organic matter only.	All oxidizable organic compounds (biodegradable + non-biodegradable) + oxidizable inorganics.
Duration of Test	Slow; standard incubation takes **5 days**.	Fast; reflux digestion takes **2 to 3 hours**.
Oxidizing Agent	Aerobic microorganisms / bacteria.	Strong chemical oxidizer ($K_2Cr_2O_7$ in boiling H_2SO_4).
Numerical Value	Always lower: $BOD < COD$.	Always higher: $COD > BOD$ (Ratio $BOD/COD \approx 0.4–0.8$ indicates good biodegradability).

4. Comprehensive Control of Pollution

A. Air Pollution Control Technologies

Particulate Matter (PM) Abatement

- **Electrostatic Precipitator (ESP):** Flue gas passes between high-voltage discharge wires (-40 to -60 kV) and grounded collection plates. Corona discharge ionizes gas; electrons charge dust particles which migrate to collection plates under electrostatic field. Collection efficiency > 99% down to 0.1 μm particles.
- **Cyclone Separator:** Uses centrifugal force to throw coarse particulate matter (> 10 μm) against outer conical walls, sliding into bottom collection hopper.
- **Fabric Filters / Baghouses:** Dirty gas passes through woven/felted fabric bags. Highly efficient (99.9%) for fine particles.
- **Wet Scrubbers (Venturi Scrubber):** Gas contacted with high-velocity liquid atomization; captures particulates and soluble acidic gases.

Gaseous Emission Abatement

- **Flue Gas Desulfurization (FGD) for SO_2:** Wet limestone scrubbing: $CaCO_3 + SO_2 \rightarrow CaSO_3 + CO_2$ followed by forced oxidation: $CaSO_3 + ½O_2 + 2H_2O \rightarrow CaSO_4{\cdot}2H_2O$ (synthetic gypsum byproduct).
- **Selective Catalytic Reduction (SCR) for NO_x:** Anhydrous ammonia (NH_3) is injected over V_2O_5-WO_3/TiO_2 catalyst at 300–400°C: $4NO + 4NH_3 + O_2 \rightarrow 4N_2 + 6H_2O$.
- **Three-Way Catalytic Converter (Automotive):** Honeycomb cordierite coated with Pt-Pd-Rh. Concurrently reduces NO_x to N_2 and oxidizes CO and unburnt hydrocarbons (HC) to CO_2 and H_2O.

B. Industrial & Municipal Wastewater Treatment Systems

- **Primary Treatment (Physical):** Bar screens remove large debris, grit chambers settle heavy mineral sand, primary sedimentation tanks remove settleable suspended organic solids (sludge settled by gravity; skimming removes grease and oil). Reduces BOD by 25–35% and TSS by 60%.
- **Secondary Treatment (Biological):** Aerobic decomposition of dissolved organic matter:
  - *Activated Sludge Process (ASP):* Wastewater enters aeration tanks supplied with compressed air. Suspended microbial flocs bio-oxidize organic carbon to CO_2 and biomass. Fluid moves to secondary clarifier where activated sludge settles; a portion is recycled (RAS) to maintain biomass inventory.
  - *Trickling Filters:* Wastewater sprayed over a packed bed of rocks or corrugated plastic media coated with microbial biofilm (zoogloeal film).
- **Tertiary / Advanced Treatment (Physicochemical Polishing):** Coagulation-flocculation using Alum ($Al_2(SO_4)_3$); granular activated carbon adsorption for non-biodegradable refractory organics; Reverse Osmosis (RO) and nanofiltration for dissolved solids; chemical disinfection via chlorination ($HOCl$) or UV irradiation to eradicate pathogenic coliform bacteria.

5. Depletion of the Stratospheric Ozone Layer

A. Stratospheric Ozone & Natural Chapman Cycle

The ozone layer located in the stratosphere (altitude 15–35 km) shields the biosphere by absorbing lethal solar Ultraviolet-B (280–315 nm) and UV-C (< 280 nm) radiation. In a pristine stratosphere, ozone concentration is maintained in dynamic photochemical equilibrium via the **Chapman Mechanism**:

CHAPMAN PHOTOCHEMICAL STEADY-STATE CYCLE

1. **Photodissociation of O_2:** $O_2 + h\nu\ (\lambda < 242\ nm) \rightarrow O(^3P) + O(^3P)$
2. **Ozone Formation:** $O + O_2 + M \rightarrow O_3 + M$ (where M is an inert third body: N_2 or O_2 absorbing excess kinetic energy)
3. **Ozone Photolysis:** $O_3 + h\nu\ (\lambda \approx 200\text{–}310\ nm) \rightarrow O_2 + O(^1D)$
4. **Recombination Sink:** $O_3 + O \rightarrow 2O_2$

B. Catalytic Depletion by Chlorofluorocarbons (CFCs / Freons) & Halons

Synthetic chlorofluorocarbons (e.g., $CFCl_3$ [CFC-11], CF_2Cl_2 [CFC-12]) are exceptionally stable and chemically inert in the troposphere. Over decades, they diffuse across the tropopause into the stratosphere, where intense shortwave UV radiation cleaves the carbon-chlorine bond, releasing atomic chlorine free radicals ($Cl^\bullet$):

FREE RADICAL CATALYTIC CHAIN REACTION

Photolytic Initiation: $CF_2Cl_2 + h\nu\ (\lambda < 220\ nm) \rightarrow CF_2Cl^\bullet + Cl^\bullet$
Catalytic Chain Propagation:

Step 1: $Cl^\bullet + O_3 \rightarrow ClO^\bullet + O_2$

Step 2: $ClO^\bullet + O \rightarrow Cl^\bullet + O_2$

Net Catalytic Destruction: $O_3 + O \rightarrow 2O_2$

*Because the active chlorine radical ($Cl^\bullet$) is fully regenerated in Step 2, a single chlorine atom can catalytically destroy over **100,000 ozone molecules** before being sequestered into stable reservoir species (HCl or $ClONO_2$).*

C. The Antarctic Ozone Hole & Polar Stratospheric Clouds (PSCs)

During the polar winter, a stable circulating wind pattern called the **polar vortex** isolates the Antarctic stratosphere. Temperatures plunge below -80°C, forming **Type I and II Polar Stratospheric Clouds (PSCs)** composed of nitric acid trihydrate ($HNO_3 \cdot 3H_2O$) and water ice crystals.

- **Heterogeneous Surface Reactions:** PSC surfaces catalytically convert benign reservoir species into highly photolabile molecular chlorine:

$$HCl(s) + ClONO_2(g) \xrightarrow{\text{[PSC surface]}} Cl_2(g) + HNO_3(s)$$
$$ClONO_2(g) + H_2O(s) \xrightarrow{\text{[PSC surface]}} HOCl(g) + HNO_3(s)$$

- **Springtime Release:** When sunlight returns to the South Pole in September/October, UV radiation rapidly photodissociates molecular chlorine ($Cl_2 + h\nu \rightarrow 2Cl^\bullet$), causing massive, localized ozone depletion (> 50% total column drop below 220 Dobson Units — the classic *Ozone Hole*).

D. Environmental Impacts & International Treaties

- **Biological & Human Consequences:** Elevated UV-B radiation damages DNA through pyrimidine dimer formation, causing non-melanoma and melanoma skin cancers; cortical cataracts and photokeratitis in eyes; immunosuppression; inhibits photosynthesis and damages cellular DNA in marine phytoplankton (the primary base of ocean trophic webs); reduces agricultural yields in sensitive crops (soybeans, wheat).
- **Mitigation & Protocols:**
  - **Montreal Protocol (1987):** Globally binding international treaty phasing out production and consumption of ozone-depleting substances (ODSs: CFCs, halons, carbon tetrachloride).
  - **Transition Substitutes:** Hydrochlorofluorocarbons (HCFCs, transitional) replaced by Hydrofluorocarbons (HFCs, non-ozone depleting).

- **Kigali Amendment (2016):** Phasing down potent greenhouse gas HFCs in favor of Hydrofluoroolefins (HFOs) and natural refrigerants (CO_2, NH_3, propane).

Quick Exam Reference Summary: All syllabus topics mandated for B.Tech Unit 3/4 Engineering Chemistry (Electrochemistry: Dry Cell, Fuel Cells, Solar Cell, DSSC; Nanochemistry: Synthesis, Classification, Applications; Environmental: Gaseous effluent analysis [NOx, SOx, H2S], Liquid effluent analysis [DO, BOD, COD], Pollution Control [Air & Water], and Ozone Layer Depletion) are systematically detailed with exact reactions, formulas, and operational parameters.